

Problem Set 3

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Contents

Problem 1	2
(a)	2
(b)	2
(c)	2
Problem 2	3

Problem 1:

(a) Let G be a group of order 27. Prove that G has an element of order 3.

Solution:

(b) Let G be a group of even order. Prove that G has an odd number of elements of order 2.

Solution:

(c) Let G be a group of order 12. Prove that G has an element of order 2.

Solution:

Problem 2: *The Twice-A-Prime Theorem*

Here we will prove a useful lemma about groups of order $2p$, where p is an odd prime — for example, groups of order 6, 10, 14, 22, $\dots$